

Homework 1 Solutions

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STAT 6530

January 31, 2026

Problem 1

Consider the function $g(y)$ defined as:

$$g(y) = \begin{cases} y^2, & \text{for } 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(i) Is $g(y)$ a pdf?

To check if the function is a pdf, we integrate it over its support to check if it equals 1.

$$\int_0^1 y^2 dy = \left[\frac{y^3}{3} \right]_0^1 = \frac{1}{3} - 0 = \frac{1}{3}$$

Since $\frac{1}{3} \neq 1$, $g(y)$ is not a pdf.

(ii) Find the constant c so that $cg(y)$ is a pdf. Denote this pdf by $f(y)$.

We need $\int_0^1 c \cdot y^2 dy = 1$. Since we know that $\int_0^1 y^2 dy = \frac{1}{3}$, we can take c to be 3. Thus, the pdf is:

$$f(y) = 3y^2, \quad 0 < y < 1$$

(iii) Show that the cdf is given by $F(y) = y^3$ for $0 < y < 1$.

The cdf is defined as $F(y) = \int_{-\infty}^y f(t) dt$. For $0 < y < 1$:

$$F(y) = \int_0^y 3t^2 dt = [t^3]_0^y = y^3$$

(iv) Finding the inverse function $F^{-1}(p)$ and the three percentiles.

$p = F(y) = y^3$. Solving for y :

$$y = p^{1/3}$$

Thus, $F^{-1}(p) = p^{1/3}$ for $0 < p < 1$.

The percentiles are calculated as follows:

- $p = 0.25 = (0.25)^{1/3} \approx 0.62996$
- $p = 0.50 = (0.50)^{1/3} \approx 0.79370$
- $p = 0.75 = (0.75)^{1/3} \approx 0.90856$

(v) Simulation using Inverse CDF method in R

```
1 set.seed(123)
2 # Generate uniform random variables
3 random_var <- runif(10000)
4 # Inverse CDF transformation
5 inverse_f <- random_var^(1/3)
6
7 # Calculate sample quartiles
8 sample_quartiles <- quantile(inverse_f, probs = c(0.25, 0.50, 0.75))
9 print(sample_quartiles)
```

Using this R code, we obtain the values: 0.6323802, 0.7908156, and 0.9058849, respectively. We see that the sample quartiles from the simulation are very close to the theoretical values calculated.

(vi) Relative Frequency Histogram

```
1 hist(inverse_f, prob=TRUE, main="Histogram of Simulated Y vs PDF of f(y)",
2       xlab="y", col="lightblue", breaks=50)
3 curve(3*x^2, from=0, to=1, col="red", lwd=2, add=TRUE)
4 legend("topleft", legend="Theoretical PDF 3y^2", col="red", lwd=2)
```

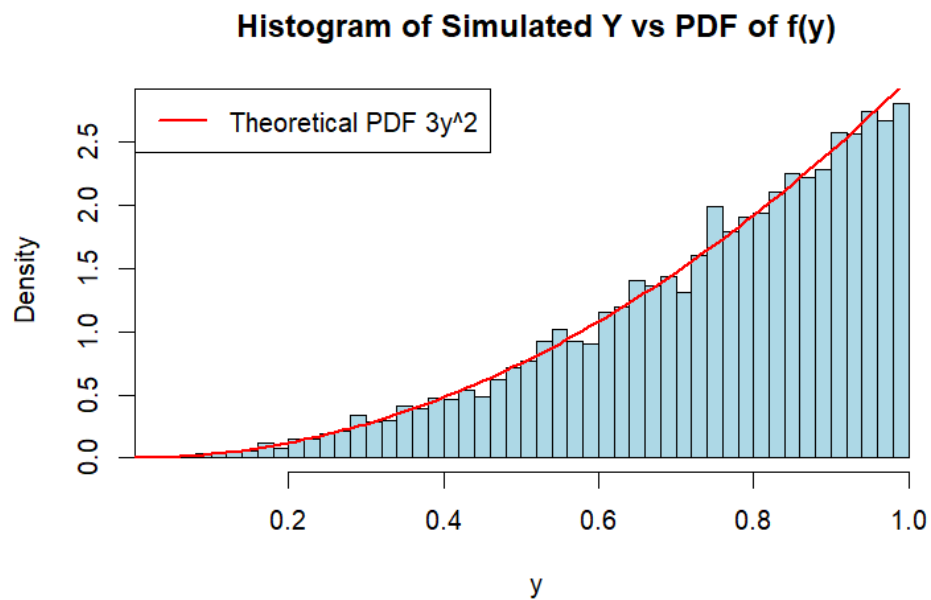


Figure 1: Relative frequency histogram of simulated data overlaid with PDF of $f(y)$.

The relative frequency histogram of the simulated data closely matches the pdf curve $f(y) = 3y^2$, showing an exponential increase in density as y approaches 1.

Problem 2

Suppose Y is a continuous random variable with the pdf $f_Y(y)$ and the cdf $F_Y(y)$. For any number c , $R(c) = \int_c^\infty f_Y(t)dt = 1 - F_Y(c)$. Consider the random variable $R(Y) = 1 - F_Y(Y)$.

(a) Find the cdf of $R(Y)$.

Let $F_{R(Y)}(r)$ denote the cumulative distribution function of the random variable $R(Y)$. By definition:

$$F_{R(Y)}(r) = P(R(Y) \leq r)$$

Substituting the expression for $R(Y)$ given in the problem:

$$F_{R(Y)}(r) = P(1 - F_Y(Y) \leq r)$$

Rearranging the inequality:

$$F_{R(Y)}(r) = P(F_Y(Y) \geq 1 - r)$$

We know that for any continuous random variable Y , the transformation $F_Y(Y)$ follows a standard Uniform distribution on the interval $(0, 1)$. For a Uniform(0,1) variable, the probability of being greater than or equal to some value k is $1 - k$ (provided $0 < k < 1$). Therefore, substituting $1 - r$ for k :

$$F_{R(Y)}(r) = 1 - (1 - r) = r$$

Thus, the cdf of $R(Y)$ is:

$$F_{R(Y)}(r) = \begin{cases} 0, & r \leq 0 \\ r, & 0 < r < 1 \\ 1, & r \geq 1 \end{cases}$$

This is a uniform distribution itself.

(b) Find the pdf of $R(Y)$.

Let $f_{R(Y)}(r)$ denote the probability density function of $R(Y)$. The pdf is the derivative of the cdf with respect to r :

$$f_{R(Y)}(r) = \frac{d}{dr}F_{R(Y)}(r) = \frac{d}{dr}(r) = 1$$

Thus, the pdf is:

$$f_{R(Y)}(r) = \begin{cases} 1, & 0 < r < 1 \\ 0, & \text{elsewhere} \end{cases}$$

This also indicates that $R(Y)$ follows a Uniform(0,1) distribution.

(c) Find the mean and the variance of $R(Y)$.

Using the pdf, we can calculate the expected value (mean) and variance directly.

Mean:

$$E[R(Y)] = \int_{-\infty}^{\infty} r \cdot f_{R(Y)}(r) dr = \int_0^1 r \cdot 1 dr = \left[\frac{r^2}{2} \right]_0^1 = \frac{1}{2}$$

Variance: First, we calculate the second moment $E[R(Y)^2]$:

$$E[R(Y)^2] = \int_0^1 r^2 \cdot f_{R(Y)}(r) dr = \int_0^1 r^2 dr = \left[\frac{r^3}{3} \right]_0^1 = \frac{1}{3}$$

Now, using the variance formula $Var(R(Y)) = E[R(Y)^2] - (E[R(Y)])^2$:

$$Var(R(Y)) = \frac{1}{3} - \left(\frac{1}{2} \right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

Problem 3

Suppose Y is a standard normal random variable with cdf $\Phi(y) = P[Y \leq y] = \int_{-\infty}^y \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$.

(a) Show that for any c , $\Phi(c) + \Phi(-c) = 1$.

By definition, $\Phi(c) = P(Y \leq c)$ and $\Phi(-c) = P(Y \leq -c)$. Because the standard normal probability density function $\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$ is symmetric about 0 (because it has a distribution $N(0, 1)$), the area under the curve to the left of $-c$ is equal to the area under the curve to the right of c . Mathematically:

$$P(Y \leq -c) = P(Y \geq c)$$

We also know that the total probability is 1:

$$P(Y \leq c) + P(Y \geq c) = 1$$

Substituting $\Phi(-c)$ for $P(Y \geq c)$ (using the symmetry property derived above):

$$\Phi(c) + \Phi(-c) = 1$$

(b) Let $X = \Phi(-Y)$. Find the cdf of X and the pdf of X .

First, we simplify the expression for X using the identity derived in part (a):

$$X = \Phi(-Y) = 1 - \Phi(Y)$$

To find the cdf $F_X(x) = P(X \leq x)$, we substitute our simplified X :

$$F_X(x) = P(1 - \Phi(Y) \leq x)$$

Rearranging the inequality:

$$F_X(x) = P(\Phi(Y) \geq 1 - x)$$

It is a known property that for any continuous random variable Y with cdf Φ , the random variable $\Phi(Y)$ follows a standard Uniform distribution on $(0, 1)$. For a standard uniform variable, the probability of being greater than or equal to a value k is simply $1 - k$. Here, $k = 1 - x$. Therefore:

$$F_X(x) = 1 - (1 - x) = x$$

Thus, the cdf is:

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

To find the pdf $f_X(x)$, we take the derivative of the cdf with respect to x :

$$f_X(x) = \frac{d}{dx} F_X(x) = \frac{d}{dx} (x) = 1$$

Thus, the pdf is:

$$f_X(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

(c) Find the mean and the variance of X .

We see that the distribution is a $U(0, 1)$ distribution, and we have already calculated the mean and variance in the last problem.

Mean: $\frac{1}{2}$

Variance: $\frac{1}{12}$

Problem 4

Consider two random variables X and Y .

(a) Show that $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$.

By definition of variance:

$$Var(X + Y) = E[((X + Y) - E[X + Y])^2]$$

Let $\mu_x = E[X]$ and $\mu_y = E[Y]$. Then $E[X + Y] = \mu_x + \mu_y$.

$$\begin{aligned} Var(X + Y) &= E[((X - \mu_x) + (Y - \mu_y))^2] \\ &= E[(X - \mu_x)^2 + (Y - \mu_y)^2 + 2(X - \mu_x)(Y - \mu_y)] \\ &= E[(X - \mu_x)^2] + E[(Y - \mu_y)^2] + 2E[(X - \mu_x)(Y - \mu_y)] \\ &= Var(X) + Var(Y) + 2Cov(X, Y) \end{aligned}$$

(b) Show that $Var(X - Y) = Var(X) + Var(Y) - 2Cov(X, Y)$.

Using the result from (a) for random variables X and $-Y$:

$$Var(X + (-Y)) = Var(X) + Var(-Y) + 2Cov(X, -Y)$$

We know that $Var(-Y) = (-1)^2 Var(Y) = Var(Y)$ and $Cov(X, -Y) = -Cov(X, Y)$. Substituting these in:

$$Var(X - Y) = Var(X) + Var(Y) - 2Cov(X, Y)$$

(c) Show how (a) and (b) simplify when X and Y are uncorrelated.

If X and Y are uncorrelated, then $Cov(X, Y) = 0$. The equations simplify to:

$$Var(X + Y) = Var(X) + Var(Y)$$

$$Var(X - Y) = Var(X) + Var(Y)$$

Problem 5

Suppose Y has an exponential distribution with pdf $f(y; \lambda) = \lambda e^{-\lambda y}$ and cdf $F(y; \lambda) = 1 - e^{-\lambda y}$.

(a) Find the median of Y .

At the median, $F(y) = 0.5$:

$$1 - e^{-\lambda y} = 0.5 \implies e^{-\lambda y} = 0.5$$

Take $\ln$ of both sides:

$$-\lambda y = \ln(0.5) = -\ln(2) \implies y = \frac{\ln(2)}{\lambda} \approx \frac{0.693}{\lambda}$$

(b) Find the lower quartile and upper quartile.

Lower Quartile (Q_1): Set $F(q_1) = 0.25$.

$$1 - e^{-\lambda q_1} = 0.25 \implies e^{-\lambda q_1} = 0.75$$

$$q_1 = \frac{-\ln(0.75)}{\lambda} = \frac{\ln(4/3)}{\lambda} \approx \frac{0.288}{\lambda}$$

Upper Quartile (Q_3): Set $F(q_3) = 0.75$.

$$1 - e^{-\lambda q_3} = 0.75 \implies e^{-\lambda q_3} = 0.25$$

$$q_3 = \frac{-\ln(0.25)}{\lambda} = \frac{\ln(4)}{\lambda} \approx \frac{1.386}{\lambda}$$

(c) Comparing median and upper quartile with the mean.

The mean of an exponential distribution is $E[Y] = \frac{1}{\lambda}$.

- **Median vs Mean:** $\frac{0.693}{\lambda} < \frac{1}{\lambda}$. The median is smaller than the mean.
- **Upper Quartile vs Mean:** $\frac{1.386}{\lambda} > \frac{1}{\lambda}$. The upper quartile is larger than the mean.

Problem 6

In an exit poll, $n = 1648$, $\hat{p} = 0.515$ for Candidate A.

(a) Find the standard error assuming $p = 0.50$.

The standard error of the sample proportion is $\sqrt{\frac{p(1-p)}{n}}$. Using $p = 0.50$:

$$SE = \sqrt{\frac{0.50(1 - 0.50)}{1648}} = \sqrt{\frac{0.25}{1648}} \approx 0.0123$$

(b) Simulation Results

(i) Simulation Code (R):

```
1 set.seed(123)
2 n_trials <- 100000
3 n_voters <- 1648
4 # Simulate election under assumption p = 0.5
5 simulations <- rbinom(n_trials, n_voters, 0.5) / n_voters
6
7 # Calculate probability of observing 0.515 or higher
8 prob_extreme <- mean(simulations >= 0.515)
9 print(paste("Probability of observing >= 51.5%:", prob_extreme))
```

We get the following results: $P(X \geq 51.5\%) = 0.11336$. We see that there is a 11.336% chance of this result.

(ii) **Using the Standard Error** We know that for a normal distribution, approximately 95% of the data lies within 2 standard deviations (standard errors) of the mean. We calculate this "non-surprising" range around the assumed population mean of 0.50:

$$\text{Range} = 0.50 \pm 2(SE)$$

$$\text{Range} = 0.50 \pm 2(0.0123) = 0.50 \pm 0.0246$$

$$\text{Interval} = [0.4754, 0.5246]$$

The observed exit poll proportion of 0.515 falls comfortably inside this interval. Therefore, obtaining a result of 51.5% when the true population is split 50/50 is not surprising, because the result is within the range of expected variability. Hence, it would be difficult to justify predicting the results of this election based on this poll.

Problem 7

Problem 7

Consider a sequence $X_1, X_2, \dots$ of independent $Ber(p)$ random variables. Let $T_n = \frac{1}{n} \sum_{i=1}^n X_i$. We define the logit function $g(x) = \log\left(\frac{x}{1-x}\right)$. We want to find the asymptotic distribution of:

$$\sqrt{n}[g(T_n) - g(p)]$$

Since each $X_i \sim Ber(p)$, we have $E[X_i] = p$ and $Var(X_i) = p(1-p)$. According to the Central Limit Theorem, the standardized quantity converges to a standard normal distribution:

$$\sqrt{n}(T_n - p) \xrightarrow{d} N(0, p(1-p))$$

We apply the Delta Method, which states that for a sequence of random variables T_n satisfying $\sqrt{n}(T_n - p) \xrightarrow{d} N(0, \sigma^2)$, the transformed variable converges as follows:

$$\sqrt{n}(g(T_n) - g(p)) \xrightarrow{d} N(0, [g'(p)]^2 \sigma^2)$$

We must find the derivative $g'(p)$. Rewriting the function using logarithm rules simplifies the differentiation:

$$g(x) = \log\left(\frac{x}{1-x}\right) = \log(x) - \log(1-x)$$

Differentiating with respect to x :

$$g'(x) = \frac{d}{dx}(\log(x)) - \frac{d}{dx}(\log(1-x)) = \frac{1}{x} - \left(\frac{1}{1-x} \cdot (-1)\right) = \frac{1}{x(1-x)}$$

Evaluated at p , we have $g'(p) = \frac{1}{p(1-p)}$.

Finally, we compute the asymptotic variance by substituting $g'(p)$ and $\sigma^2 = p(1-p)$ into the Delta Method formula:

$$Var_{asympt} = [g'(p)]^2 \cdot \sigma^2 = \left[\frac{1}{p(1-p)}\right]^2 \cdot p(1-p)$$

$$Var_{asympt} = \frac{p(1-p)}{p^2(1-p)^2} = \frac{1}{p(1-p)}$$

Thus, the final convergence in distribution is:

$$\sqrt{n} \left[\log\left(\frac{T_n}{1-T_n}\right) - \log\left(\frac{p}{1-p}\right) \right] \xrightarrow{d} N\left(0, \frac{1}{p(1-p)}\right)$$

Problem 8

Using Chebyshev's inequality to prove the law of large numbers.

Chebyshev's inequality states that if X is a random variable with finite variance σ^2 and mean μ , then for any $k > 0$:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with mean μ and finite variance σ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. The expected value and variance of the sample mean are:

$$E[\bar{X}_n] = \mu$$

$$Var(\bar{X}_n) = \frac{\sigma^2}{n}, \quad \text{so } SD(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

We want to show that for any $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq \epsilon) = 0$$

Applying Chebyshev's inequality to the random variable $Y = \bar{X}_n$. We set the distance ϵ equal to k standard deviations:

$$\epsilon = k \cdot SD(\bar{X}_n) = k \frac{\sigma}{\sqrt{n}}$$

Solving for k :

$$k = \frac{\epsilon\sqrt{n}}{\sigma}$$

Substituting this into Chebyshev's bound ($\frac{1}{k^2}$):

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{1}{\left(\frac{\epsilon\sqrt{n}}{\sigma}\right)^2} = \frac{\sigma^2}{n\epsilon^2}$$

Taking the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{\sigma^2}{n\epsilon^2} = 0$$

Since probabilities cannot be negative, the probability converges to 0. This proves the Law of Large Numbers.